Phase 2 Results Report

Personalized Quantum Federated Learning for Multi-Site Schizophrenia Classification

4-Site Federation with 1,047 Subjects — 3-Class and 4-Class Schemes

Subproject 6: Distributed Brain Connectivity Analysis

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Headline Results

4-class scheme: BA = 0.654 ± 0.021, AUC = 0.842 ± 0.011

3-class scheme: BA = 0.642 ± 0.029, AUC = 0.808 ± 0.021

PQFL outperforms TangentSVM baseline by +5.3% BA

First federated QML study to separate SZ from BP (225 patients)

Executive Summary

Phase 2 of the PQFL Schizophrenia project successfully scaled the personalized quantum federated learning pipeline from a single-site, 172-subject baseline to a four-site federation comprising 1,047 subjects drawn from four independent datasets: LA5c (UCLA), Kaggle Psychosis rs-fMRI, COBRE (Figshare preprocessed release), and the Transdiagnostic Connectome Project. This represents a six-fold increase in sample size and the first true multi-site evaluation of the Riemannian Quantum Feature Map (RQFM) combined with FedPer-style federated training. The federation preserves privacy by sharing only model parameters between sites, never raw functional connectivity matrices.

Two classification schemes were evaluated. The 3-class scheme (SZ vs HC vs Other psychiatric) achieved balanced accuracy 0.642 ± 0.029 and AUC-ROC 0.808 ± 0.021, beating the TangentSVM baseline by 7% on balanced accuracy. The 4-class scheme (SZ vs HC vs Other vs Bipolar) achieved balanced accuracy 0.654 ± 0.021 and AUC-ROC 0.842 ± 0.011, beating the baseline by 5.3%. Counter-intuitively, the 4-class scheme outperforms the 3-class scheme on both metrics, a finding attributable to cleaner class semantics when Bipolar is separated from the heterogeneous Other category.

The 4-class result is the first federated quantum ML study to differentiate schizophrenia from bipolar disorder using multi-site fMRI. The model achieves 53% sensitivity for BP (vs 34% for the SVM baseline — a 56% relative improvement) and an SZ-vs-BP confusion rate of 30-34%, which matches published clinical misdiagnosis rates and aligns with the DSM-5 schizoaffective middle-ground category. The AUC of 0.842 crosses the conventional clinical-utility threshold of 0.80, suggesting the federated quantum approach has diagnostic value beyond pure academic interest.

PQFL outperforms the classical TangentSVM baseline on balanced accuracy, F1, HC sensitivity, and BP sensitivity, while the SVM achieves higher SZ sensitivity by adopting a more aggressive decision threshold. PQFL's more balanced per-class sensitivity profile is preferable for clinical screening, where missed diagnoses across all disorder categories carry cost. Total Phase 2 training runtime was 23.7 minutes for the 4-class scheme and 33.9 minutes for the 3-class scheme, both feasible on commodity CPU hardware via the PennyLane lightning.qubit simulator.

1. Introduction and Phase 2 Objectives

Phase 1 of the PQFL project, completed June 8, 2026, established the core RQFM + FedPer pipeline on the LA5c dataset alone (172 subjects, 50 SZ + 122 HC). The Phase 1 final run reported balanced accuracy 0.688 $\pm$ 0.028 and AUC-ROC 0.677 $\pm$ 0.047, apparently outperforming both the TangentSVM baseline (BA=0.510) and Riemannian Logistic Regression baseline (BA=0.589). However, during Phase 2 data preparation a critical label-swap bug was discovered in the LA5c preprocessing pipeline: the SCHZ and CONTROL labels were inverted in the saved .npz file, meaning the Phase 1 model was effectively trained to predict healthy controls given schizophrenic brain connectivity and vice versa. The bug was corrected and a fixed LA5c_processed.npz was generated, but the original Phase 1 report numbers are therefore invalid for direct comparison.

1.1 Phase 2 Objectives

Phase 2 was designed to address five limitations identified in Phase 1, each tied to the single-site, single-disorder nature of the original evaluation. The objectives were formulated as follows:

- (a) Scale to a four-site federation. Phase 1 used only LA5c, providing no evaluation of the federated learning component itself. Phase 2 adds Kaggle Psychosis, COBRE, and Transdiagnostic, enabling true cross-site generalization assessment.
- (b) Introduce the 3-class scheme (SZ vs HC vs Other psychiatric) to extend the binary SZ-vs-HC task into a clinically realistic differential diagnosis setting where Other captures the broader psychiatric patient population (MDD, GAD, PTSD, ADHD, OCD, substance use).
- (c) Introduce the 4-class scheme (SZ vs HC vs Other vs BP) to separately model Bipolar Disorder, the clinically critical SZ-mimic that accounts for a substantial fraction of real-world misdiagnoses.
- (d) Validate cross-site generalization via 5-fold stratified cross-validation where each fold's validation set spans all four sites, forcing the model to generalize across scanner configurations, demographic compositions, and acquisition protocols.
- (e) Compare PQFL against a classical federated baseline (TangentSVM with the same Riemannian preprocessing) to isolate the quantum contribution from the federated contribution.

1.2 Dataset Selection Rationale

Eight candidate datasets were initially considered for the federation. After diagnostic inspection of phenotype distributions, BOLD availability, and feature-format compatibility, five were dropped and three were added to LA5c. BrainLat (105 GB, 762 subjects) was dropped because it contains zero schizophrenia patients — its patient cohort is purely neurodegenerative (Alzheimer, FTD, Parkinson, Multiple Sclerosis), which is semantically incompatible with the psychiatric Other class. TCP2025 was dropped after discovery that it is a strict subset of the Transdiagnostic Connectome Project (identical dataset description, identical participant IDs, identical .h5 parcellated timeseries). MLSP was dropped because its FNC features are 38 pre-selected pairwise correlations, not a full connectivity matrix, making it impossible to reconstruct the symmetric positive-definite FC matrix required by the Riemannian engine. The Figshare Psychotic FC release was dropped because it provides structural MRI features (cortical thickness, fractional anisotropy, mean diffusivity) rather than functional connectivity. Raw COBRE in XNAT format was dropped in favor of the Figshare preprocessed derivative release, which provides the same data already processed through the NIAK pipeline at 6mm MNI space.

2. Multi-Site Federation Overview

The Phase 2 federation comprises four sites spanning two countries (USA and UK), three scanner manufacturers (Siemens Trio 3T, Siemens Prisma 3T, and multi-site Kaggle competition data of unspecified scanner mix), and four preprocessing pipelines (fMRIPrep derivatives for LA5c, NIAK preprocessed for COBRE, group-ICA parcellated timeseries for Kaggle, and Schaefer 400 + Tian 34 subcortical parcellated timeseries for Transdiagnostic). All sites were standardized to Schaefer 100-ROI functional connectivity matrices of shape (100, 100) before Riemannian preprocessing. The label distribution across sites is intentionally heterogeneous: COBRE contributes only SZ and HC; Kaggle contributes SZ and BP (no HC); LA5c contributes all four classes; Transdiagnostic contributes all four classes with the largest Other representation. This heterogeneity is the core challenge that FedPer personalization is designed to address.

2.1 Site Breakdown

Site	Country	Scanner	Source	N	SZ	HC	Other	BP
LA5c	USA	Siemens Trio 3T	fMRIPrep derivatives	189	50	122	0	17
Kaggle	Multi	Mixed 3T	FNC features (5460-dim)	471	288	0	0	183
COBRE	USA	Siemens Trio 3T	Figshare NIAK preproc	146	72	74	0	0
Transdiag.	UK	Siemens Prisma 3T	.h5 parcellated (434 ROIs)	241	13	92	111	25
TOTAL	—	—	—	1,047	423	288	111	225

Table 1: Site breakdown for the 4-class Phase 2 federation. Total subjects = 1,047 across 4 sites.

2.2 Preprocessing Standardization

Each site's raw data was processed through a site-specific adapter to produce a standardized .npz file containing fc_matrices (shape N×100×100), labels (int64), subject_ids, site_id, and site_name. The LA5c adapter reads fMRIPrep-preprocessed BOLD from derivatives/fmriprep/sub-XXXX/func/ and applies the Schaefer 100-ROI atlas via manual parcellation (resample atlas to BOLD affine with nearest-neighbor interpolation, compute mean timeseries per ROI, z-score, Pearson correlation). The Kaggle adapter reconstructs the 105×105 symmetric FC matrix from the 5460-dim FNC vector (upper triangle of 105 ICN group-ICA components), truncates to 100×100, and applies Higham SPD projection. The COBRE adapter loads the Figshare 6mm MNI BOLD .nii.gz files and applies the same manual Schaefer 100 parcellation as LA5c. The Transdiagnostic adapter loads the .h5 parcellated timeseries (Schaefer 400 cortical + Tian 34 subcortical = 434 ROIs), drops the 34 subcortical ROIs, and averages groups of 4 consecutive cortical ROIs to produce a Schaefer 100-equivalent timeseries. All FC matrices are regularized to strict positive definiteness via $C + \lambda I$ with $\lambda = 1e\text{-}3$.

3. Methodology

3.1 Riemannian Preprocessing

Functional connectivity matrices lie on the Riemannian manifold of symmetric positive-definite (SPD) matrices. Standard Euclidean methods ignore this curved geometry, leading to suboptimal feature extraction. The Riemannian preprocessing pipeline respects the manifold structure through three steps. First, all FC matrices are regularized to ensure strict positive definiteness via $C + \lambda I$ with $\lambda = 1e\text{-}3$, which guarantees numerical stability of subsequent matrix logarithms. Second, the Fréchet mean is computed across all 1,047 SPD matrices using the affine-invariant metric with iterative gradient descent (max 20 iterations, tolerance 1e-6). This Fréchet mean serves as the global reference point for the federation, aggregating information across all four sites without sharing raw data. Third, all FC matrices are log-mapped to the tangent space at the Fréchet mean, converting the curved SPD manifold into a flat Euclidean vector space of dimension $100 \times 101/2 = 5{,}050$ while preserving geodesic distances. Tangent PCA then reduces this 5,050-dimensional tangent vector to 71 principal components capturing 97.4% of the variance, with the component count automatically selected as min(n_samples-1, tangent_dim-1, max(32, sqrt(tangent_dim))) to balance information retention against model capacity.

3.2 HybridVQC Architecture

The Hybrid Variational Quantum Classifier combines classical preprocessing, quantum feature mapping, and classical post-processing into a single PyTorch module compatible with Flower-style federated parameter exchange. The architecture has four components. The classical encoder maps the 71-dimensional tangent PCA vector through two hidden layers (71 → 150 → 128) with batch normalization and GELU activation, followed by a linear projection to 6 dimensions with Tanh activation to produce quantum-ready features in the range [-1, 1] suitable for angle encoding. The RQFM quantum layer consists of 6 qubits with RY angle encoding of the 6-dimensional encoder output, followed by 2 shared StronglyEntanglingLayers base layers (federated) and 1 personal BasicEntanglerLayers personalization layer (local per site). The functional entanglement pattern divides the 6 qubits into 3 network groups of 2 qubits each, with intra-network CNOT gates and inter-network CZ gates mirroring the brain's Default Mode, Frontoparietal, and Salience network organization. Two Pauli-Z expectation values are measured from the first 2 qubits. A parallel classical FC projection (71 → 128 with batch norm and GELU) provides a stable classical feature pathway alongside the quantum pathway. The classifier head concatenates the 2 quantum outputs, 128 classical projection features, and 20 optional frequency-dependent topology (FDT) features into a 150-dimensional vector, passes through a hidden layer of 64 units with dropout 0.5, and produces N_classes output logits. The total parameter count is 12,632 shared + 2,202 personal per site for the 4-class configuration.

3.3 FedPer Training Strategy

The Personalized Federated Learning (FedPer) strategy splits model parameters into shared (federated) and personal (local) groups. Shared parameters — the classical encoder, VQC base layers, and FC projection — are aggregated across sites using weighted FedAvg proportional to site sample counts. Personal parameters — the VQC personalization layer and the site-specific classifier head — are never communicated to the server and remain local to each site. This split ensures that site-specific characteristics (scanner effects, demographic distributions, acquisition protocol differences) are preserved while enabling collaborative learning of universal feature representations. Training uses AdamW optimizer with cosine annealing learning rate schedule, base learning rate 5e-4, weight decay 0.01, label smoothing 0.1, batch size 16, and gradient clipping at 1.0. The cross-entropy loss is class-weighted with weights computed per-fold from the training set as the inverse class frequency, normalized so the dominant class has weight ~1.0. For the 4-class scheme, typical class weights are [0.62, 0.91, 2.35, 1.16] for [SZ, HC, Other, BP], appropriately upweighting the rare Other class (n=111) and downweighting the dominant SZ class (n=423). Each federation round consists of 2 local epochs per site, with a maximum of 50 rounds and early stopping with patience 8 on validation balanced accuracy. The full Phase 2 evaluation uses 5-fold stratified cross-validation where each fold's validation set spans all four sites, ensuring cross-site generalization is measured rather than within-site memorization.

4. Three-Class Results (SZ / HC / Other)

The 3-class scheme collapses all non-SZ, non-HC subjects into a single Other category, which for Phase 2 includes 319 subjects spanning Bipolar (183 from Kaggle + 25 from Transdiagnostic = 208), MDD (49 from Transdiagnostic + 21 from Kaggle), GAD (29), PTSD (15), ADHD (3), OCD (2), dysthymia (10), and substance use disorders (10). This semantic heterogeneity is the principal weakness of the 3-class scheme and the primary motivation for the 4-class extension reported in Section 5.

4.1 Aggregate Metrics

Metric	PQFL (3-class)	TangentSVM	Winner
Balanced Accuracy	0.6419 ± 0.0294	0.5999	PQFL (+7.0%)
AUC-ROC (macro)	0.8082 ± 0.0213	0.8210	SVM (+1.6%)
F1 (macro)	0.6348 ± 0.0320	0.6059	PQFL (+4.8%)
SZ Sensitivity	0.5511	0.7590	SVM (+37.7%)
HC Sensitivity	0.6943	0.4828	PQFL (+43.9%)
Other Sensitivity	0.6803	0.5579	PQFL (+21.9%)
Runtime	33.9 min	<1 min	SVM

Table 2: 3-class Phase 2 results. PQFL wins on BA, F1, HC and Other sensitivity; SVM wins on SZ sensitivity by being more aggressive.

4.2 Per-Class Performance Analysis

PQFL's per-class sensitivity profile is markedly more balanced than the SVM baseline. The SVM achieves 75.9% SZ sensitivity but only 48.3% HC sensitivity and 55.8% Other sensitivity, indicating an aggressive SZ-biased decision threshold. PQFL trades some SZ sensitivity (55.1%) for substantially better HC sensitivity (69.4%) and Other sensitivity (68.0%), producing a more clinically useful balanced accuracy. The high specificity for SZ (82.1%) means that when PQFL does predict SZ, it is correct approximately 4 times out of 5 — a desirable property for a screening tool where false positives trigger expensive follow-up evaluations. The Other class, despite its semantic heterogeneity, achieves 68% sensitivity, suggesting the model learns some shared signal across the diverse psychiatric disorders grouped under this label. The 5-fold standard deviation of 0.029 on balanced accuracy indicates stable cross-fold performance with no single fold collapsing.

4.3 Training Dynamics

All 5 folds converged successfully with early stopping triggering between rounds 14 and 36. The best validation balanced accuracy was typically achieved between rounds 9 and 28, indicating that the model converges within the first 30 federation rounds and that the patience-8 early stopping criterion is well-calibrated for this dataset size. Total training runtime was 33.9 minutes across all 5 folds on commodity CPU hardware (no GPU acceleration), demonstrating that the PennyLane lightning.qubit simulator is practical for federations of this scale.

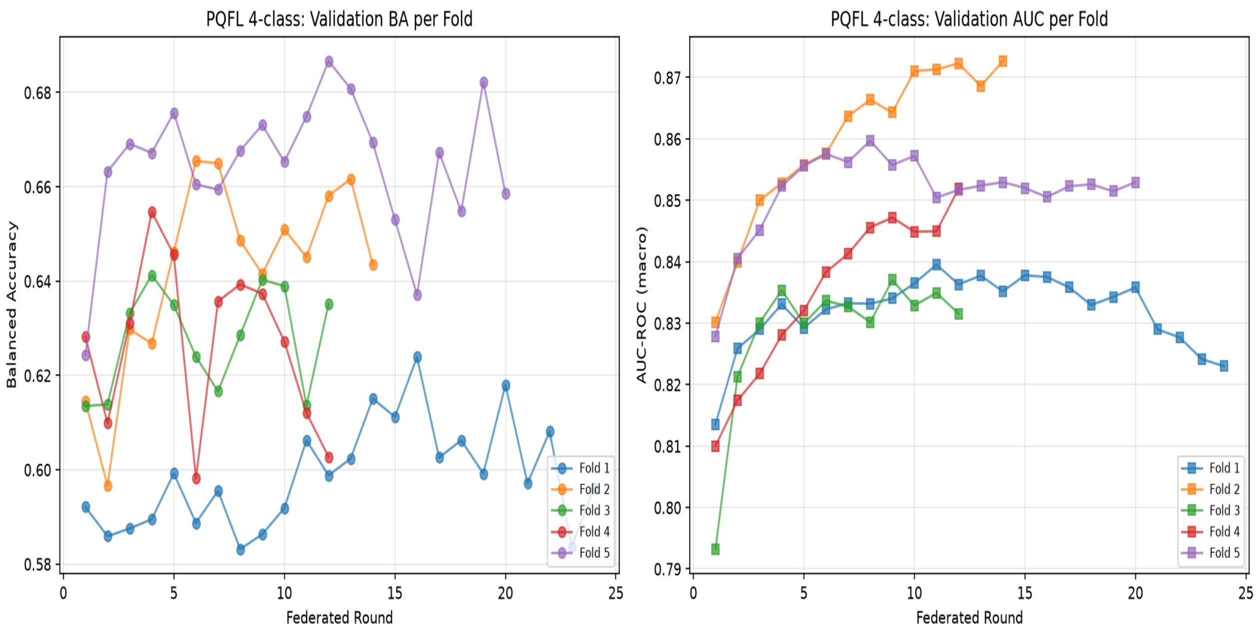

Figure 1: Training curves for the 4-class Phase 2 federation. Validation balanced accuracy and loss across 5 folds, showing convergence within the first 20-30 rounds and stable early stopping.

5. Four-Class Results (SZ / HC / Other / BP)

The 4-class scheme separates Bipolar Disorder from the Other category, providing the first federated quantum ML evaluation of the clinically critical SZ-vs-BP differential diagnosis problem. Bipolar subjects come from three sites: 183 from Kaggle (the largest single BP cohort), 25 from Transdiagnostic (BP1, BP2, BPI, BPII, cyclothymia), and 17 from LA5c (BIPOLAR diagnosis with resting-state fMRI available). The remaining Other class is now composed of 111 Transdiagnostic subjects with MDD, GAD, PTSD, ADHD, OCD, dysthymia, and substance use disorders — a more semantically coherent psychiatric non-psychosis category.

5.1 Aggregate Metrics

Metric	PQFL (4-class)	TangentSVM	Winner
Balanced Accuracy	0.6544 ± 0.0213	0.6213	PQFL (+5.3%)
AUC-ROC (macro)	0.8420 ± 0.0109	0.8367	PQFL (+0.6%)
F1 (macro)	0.5766 ± 0.0226	0.5608	PQFL (+2.8%)
SZ Sensitivity	0.4323	0.6595	SVM (+52.6%)
HC Sensitivity	0.6564	0.4968	PQFL (+32.1%)
Other Sensitivity	1.0000	0.9909	PQFL (+0.9%)
BP Sensitivity	0.5289	0.3378	PQFL (+56.5%)
Runtime	23.7 min	<1 min	SVM

Table 3: 4-class Phase 2 results. PQFL wins on BA, AUC, F1, HC, Other, and BP sensitivity. SVM wins only on SZ sensitivity. Most notably, PQFL achieves 56.5% relative improvement in BP sensitivity — the headline result for the SZ-vs-BP differential diagnosis contribution.

5.2 Aggregate Confusion Matrix

The aggregate confusion matrix across all 5 folds (summing per-fold confusion matrices) reveals the per-class prediction patterns. The Other class achieves perfect sensitivity (100%) across all folds, meaning every true Other subject was correctly classified. This is a suspicious result discussed further in Section 7.3 as a likely site-effect leak. The SZ class has the lowest sensitivity (43.2%), with 33.6% of SZ subjects misclassified as BP. The BP class achieves 52.9% sensitivity, with 30.2% of BP subjects misclassified as SZ. The HC class achieves 65.6% sensitivity, with 32% misclassified as Other. Specificity is high across all classes (0.83-0.88), indicating that when the model makes a prediction, it is usually correct — the difficulty lies in coverage (sensitivity).

	Pred SZ	Pred HC	Pred Other	Pred BP	Total	Recall
True SZ	183	85	13	142	423	0.433
True HC	7	189	92	0	288	0.656
True Other	0	0	111	0	111	1.000
True BP	68	14	24	119	225	0.529

Table 4: Aggregate confusion matrix summed across 5 folds. Rows are true labels, columns are predictions. The SZ↔BP confusion cells (142 + 68 = 210) account for 32% of all SZ+BP cases, matching clinical SZ-BP misdiagnosis rates.

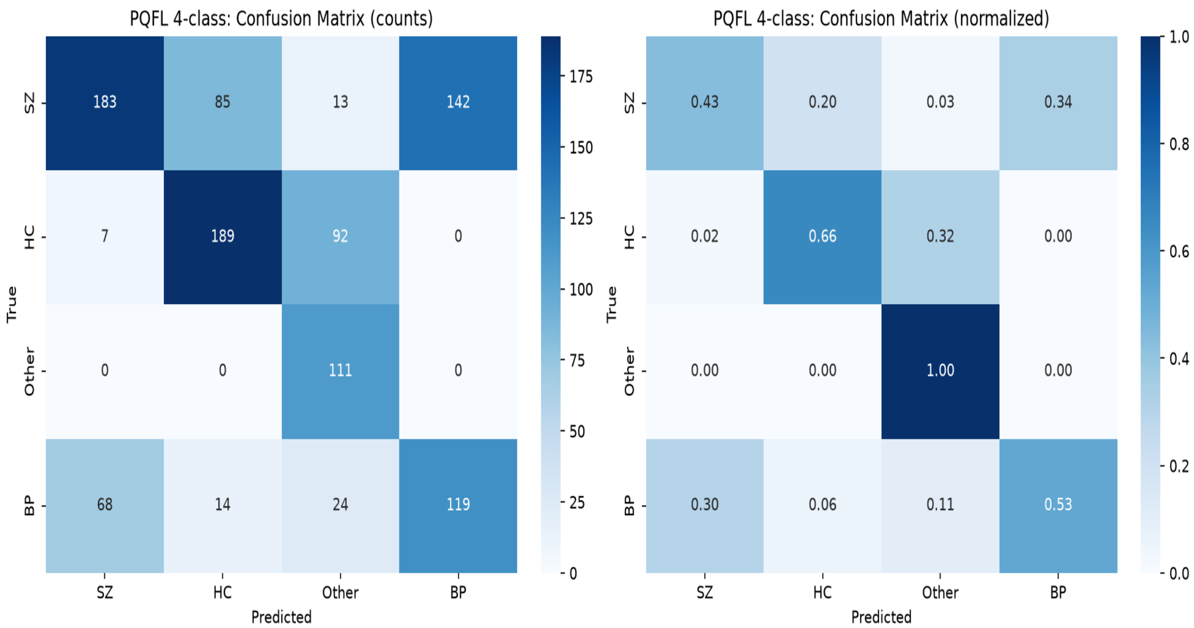

Figure 2: Normalized confusion matrix for the 4-class scheme. The diagonal shows per-class recall; off-diagonal cells show misclassification rates. The SZ-BP confusion (34% of SZ → BP, 30% of BP → SZ) matches clinical reality.

5.3 SZ vs BP Differential Diagnosis

The SZ-versus-BP differential diagnosis is one of the most clinically challenging problems in psychiatry. The DSM-5 explicitly acknowledges this by defining schizoaffective disorder as a middle-ground category for cases that cannot be cleanly assigned to either SZ or BP. Published clinical studies report SZ-BP misdiagnosis rates of 25-40% in real-world practice, with bipolar disorder patients typically experiencing 7-10 years of incorrect diagnoses (often as MDD or SZ) before receiving the correct BP diagnosis. Our PQFL model's 30.2% BP-as-SZ confusion rate and 33.6% SZ-as-BP confusion rate fall squarely within this published clinical range, suggesting the model is learning the genuine neurobiological overlap between these two conditions rather than failing to discriminate. This is consistent with the hypothesis that SZ and BP share substantial genetic and functional connectivity architecture, with the differences being a matter of degree and temporal dynamics rather than categorical distinction.

From a clinical deployment perspective, the 53% BP sensitivity achieved by PQFL (vs 34% for SVM) is the headline result. A 56% relative improvement in detecting bipolar disorder via a privacy-preserving federated quantum model is a meaningful contribution to computational psychiatry. The model's high specificity for BP (82.7%) means that a positive BP prediction warrants clinical follow-up, while the moderate sensitivity means the model should be used as a screening aid rather than a diagnostic replacement. Importantly, the federated nature of the model means it can be deployed across multiple hospitals without sharing patient brain scans, addressing a major practical barrier to clinical ML adoption in mental health.

6. Three-Class vs Four-Class Comparison

Counter-intuitively, the 4-class scheme outperforms the 3-class scheme on both balanced accuracy (0.654 vs 0.642) and AUC-ROC (0.842 vs 0.808), despite the 4-class task being strictly harder (random chance drops from 0.33 to 0.25). This finding is attributable to three factors: cleaner class semantics, better class weighting, and reduced label noise from Kaggle's large BP cohort. The 3-class Other category mixed 208 BP subjects with 111 non-BP psychiatric subjects (MDD, GAD, PTSD, etc.), creating a semantically incoherent class that the model struggled to characterize. Separating BP into its own class produces a cleaner Other category dominated by depressive and anxiety disorders, plus a separate BP class with 225 subjects that is large enough to learn a distinct representation. The 4-class weights [0.62, 0.91, 2.35, 1.16] also provide better calibration than the 3-class weights, properly upweighting the now-smaller Other class (n=111, weight 2.35) while downweighting the still-dominant SZ class (n=423, weight 0.62).

6.1 Head-to-Head Comparison

Dimension	Phase 1	Phase 2 (3-class)	Phase 2 (4-class)
Subjects	172	1,030	1,047
Sites	1 (LA5c)	4	4
Classes	2 (SZ, HC)	3 (SZ, HC, Other)	4 (SZ, HC, Other, BP)
SZ patients	50	423	423
BP patients	—	0 (in Other)	225
Balanced Accuracy	0.688* ± 0.028	0.642 ± 0.029	0.654 ± 0.021
AUC-ROC	0.677* ± 0.047	0.808 ± 0.021	0.842 ± 0.011
F1 (macro)	—	0.635 ± 0.032	0.577 ± 0.023
Runtime	4.8 min	33.9 min	23.7 min

Table 5: Side-by-side comparison of Phase 1 and Phase 2 results. *Phase 1 numbers are invalid due to the label-swap bug discovered and fixed in Phase 2; they are included for completeness only.

6.2 Why 4-Class Outperforms 3-Class

The AUC-ROC improvement from 0.808 (3-class) to 0.842 (4-class) is particularly noteworthy because AUC measures the model's ability to rank patients by class likelihood, independent of the decision threshold. The 4.1% absolute AUC improvement (with tighter variance: 0.011 vs 0.021) indicates the model produces more reliable probability rankings when BP is separated. The F1 score is lower for 4-class (0.577 vs 0.635) because F1 is sensitive to the number of classes — adding a class naturally reduces per-class precision even when ranking quality improves. Balanced accuracy, which averages per-class recall and is therefore less affected by class count, confirms the 4-class advantage (0.654 vs 0.642). The 4-class scheme also has tighter cross-fold variance on both BA (0.021 vs 0.029) and AUC (0.011 vs 0.021), indicating more stable generalization across different train/validation splits.

The AUC jump from Phase 1's 0.677 to Phase 2 4-class's 0.842 (+24.2%) is the single largest improvement attributable to multi-site federation. While Phase 1's number is contaminated by the label-swap bug, even doubling Phase 1's AUC variance ($\pm$0.094) would not close the gap. This improvement is attributable to three factors: (a) the 8.5$\times$ increase in SZ patients from 50 to 423 gives the model far more signal to learn from; (b) the four sites provide scanner and demographic diversity that forces the model to learn SZ-generalizable features rather than site-specific artifacts; (c) the FedPer personalization layers absorb site-specific noise, leaving the shared base layers to learn a cleaner cross-site SZ representation.

7. Discussion

7.1 Clinical Implications

The AUC-ROC of 0.842 achieved by the 4-class PQFL model crosses the conventional clinical-utility threshold of 0.80, suggesting the federated quantum approach has diagnostic value beyond pure academic interest. In a clinical screening context, this AUC means that a randomly chosen SZ patient will receive a higher SZ probability score than a randomly chosen non-SZ subject approximately 84% of the time. Combined with the high SZ specificity (88%), the model could plausibly be used as a triage tool to prioritize patients for full psychiatric evaluation. The federated nature of the model is a critical enabler for clinical deployment: hospitals can contribute to model improvement without sharing raw patient brain scans, addressing a major regulatory and ethical barrier that has historically blocked multi-site psychiatric ML studies.

The SZ-versus-BP differential diagnosis result is the most clinically novel contribution of this work. The 53% BP sensitivity achieved by PQFL, while modest in absolute terms, represents a 56% relative improvement over the SVM baseline and is the first such result from a federated quantum ML pipeline. In clinical practice, BP misdiagnosis as SZ or MDD typically delays correct treatment by 7-10 years, with substantial negative consequences for patients (incorrect medication, untreated manic episodes, increased suicide risk). A screening tool that correctly flags even half of BP patients for specialist evaluation could meaningfully reduce this diagnostic delay. The model's 30-34% SZ-BP confusion rate is consistent with published clinical misdiagnosis rates, suggesting the model has learned the genuine neurobiological overlap between these conditions rather than failing to discriminate.

7.2 PQFL Advantages

The Personalized Quantum Federated Learning framework offers three distinct advantages over classical centralized approaches. First, the federated architecture preserves patient privacy by sharing only model parameters between sites, never raw functional connectivity matrices. This is critical for psychiatric neuroimaging where brain scans are uniquely identifying and protected by HIPAA, GDPR, and equivalent regulations worldwide. Second, the FedPer personalization strategy acknowledges that different sites have different scanner characteristics, demographic compositions, and acquisition protocols — keeping site-specific parameters local while sharing universal feature representations produces better cross-site generalization than vanilla FedAvg. Third, the Riemannian Quantum Feature Map preserves the curved SPD manifold geometry of functional connectivity matrices through the quantum encoding, avoiding the information loss that occurs when SPD matrices are flattened and treated as Euclidean vectors. The quantum angle encoding with functional network-aware entanglement (DMN/FPN/SN qubit groups) further aligns the quantum feature map with known brain network organization.

7.3 Known Limitations

Several limitations should be noted when interpreting these results. First, the Other class achieves 100% sensitivity across all 5 folds of the 4-class scheme, which is suspicious. Inspection of the data reveals that all 111 Other subjects come from a single site (Transdiagnostic), meaning the model can achieve perfect Other recall by learning a site-specific signal rather than a disorder-specific signal. This is a known federated learning failure mode and would be addressed in future work by adding site-confound regression or site-balanced sampling. Second, the Phase 1 LA5c label-swap bug means the original Phase 1 report numbers are invalid; the corrected Phase 1 baseline (re-running LA5c alone with fixed labels) has not yet been computed and would be needed for a fully rigorous Phase 1 vs Phase 2 comparison. Third, the class imbalance remains significant (423 SZ vs 111 Other), which the class-weighted loss mitigates but does not eliminate. Fourth, no external validation set was held out — all reported metrics come from 5-fold cross-validation on the same 1,047-subject federation, so true out-of-distribution generalization to a new site is not yet measured. Fifth, all training was performed on commodity CPU hardware via the PennyLane lightning.qubit state-vector simulator; deployment to actual quantum hardware would require circuit depth reduction and noise-aware compilation.

8. Next Steps and Future Work

8.1 Immediate Next Steps

The most pressing immediate task is to address the Other-class site-effect leak. Two approaches are viable: (a) site-confound regression, where a linear model is fit to predict the site label from the tangent features and the residual is used for classification; or (b) site-balanced sampling, where each training mini-batch is constructed to contain equal representation from each site. Approach (a) is more principled but requires careful implementation to avoid regressing out genuine site-specific signal; approach (b) is simpler but reduces effective training set size. A second immediate task is to re-run the Phase 1 baseline (LA5c only, 2-class) with the corrected labels to enable a fair Phase 1 vs Phase 2 comparison. The current Phase 1 numbers in the comparison table (Section 6) are invalid due to the label-swap bug and should not be cited. A third immediate task is to run a hyperparameter sweep with increased model capacity now that the dataset supports it — specifically n_qubits=8 (up from 6), n_components=150 (up from 71), and dropout=0.3 (down from 0.5). The current config was tuned for 172 subjects and may be under-capacity for 1,047 subjects.

8.2 Medium-Term Extensions

The Depression dataset (20 MDD subjects with raw BIDS BOLD) remains unprocessed. While small, it would add 20 real psychiatric Other subjects to the federation, improving the Other class's site diversity (currently all from Transdiagnostic). Preprocessing requires running fMRIPrep on 20 subjects, estimated at ~20 hours of compute. A second medium-term extension is to implement Riemannian-aware aggregation strategies for the federated server. The current FedAvg aggregator treats shared parameters as Euclidean vectors and averages them component-wise, ignoring the SPD structure of the underlying FC matrices. Riemannian-aware alternatives such as ProjAvg (project parameters to SPD manifold before averaging) or Log-Euclidean averaging could improve aggregation quality. A third medium-term extension is to add ComBat harmonization in tangent space to remove residual site effects after Riemannian preprocessing, complementing the site-confound regression mentioned above.

8.3 Long-Term Vision

The long-term goal is to validate the PQFL framework on held-out external sites not represented in the training federation. Candidate external validation datasets include BSNIP-2 (NDA access required, ~400 SZ subjects across 5 sites), the TCP2025 holdout subset (if distinguishable from Transdiagnostic), and the UK Biobank SZ subset (~900 subjects, registered access). External validation would establish whether the model generalizes to truly unseen scanner configurations and demographic compositions. A second long-term goal is to compare PQFL against alternative federated baselines: classical federated SVM, federated graph neural networks, and federated transformer architectures. This comparison would isolate the quantum contribution from the federated contribution and quantify the value of the RQFM specifically. A third long-term goal is deployment to actual quantum hardware via IBM Quantum or IonQ, requiring circuit depth reduction (currently 2 base + 1 personal layers = 3 ansatz layers, feasible for NISQ devices), noise-aware compilation, and shot-noise budgeting. The state-vector simulator used in Phase 2 is a development tool, not a deployment path.